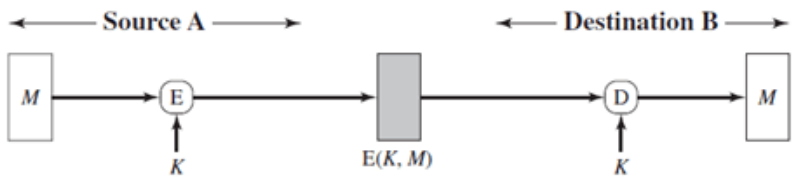
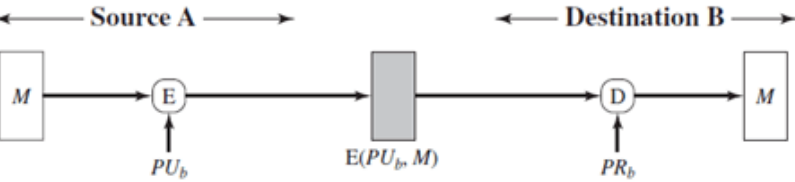
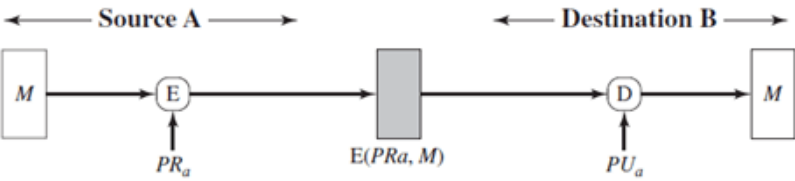
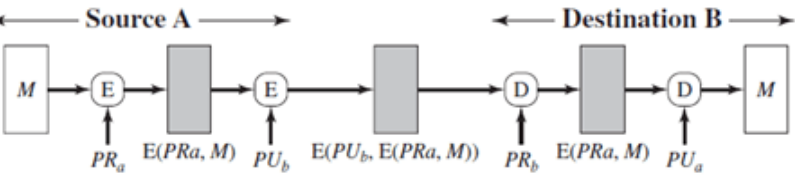


Semester: August 2021 - November 2021 Examination: In Semester Examination		
Programme code: Programme: Honours Programme in Cyber Security & Forensics	Class: SY	Semester: III(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: COMP	
Course Code:116h55C301	Name of the Course: Applied Cryptography	

Question No.		Max. Marks
Q.1	1.1) The scheme in which a random key is used to encrypt and decrypt a single message, and then is discarded is called as _____ a) Playfair Cipher b) Vernam Cipher c) Vignere Cipher d) One Time Pad	1
	1.2) Kasiski test is used for cryptanalyzing _____ cipher a) Vernam Cipher b) Vignere Cipher c) Hill Cipher d) Playfair Cipher	1
	1.3) Which of the following is False with respect to Asymmetric Key Cryptography? a) Each user generates a pair of keys b) Bob encrypts the message using Alice's public key. When Alice receives the message, she decrypts it using her private key. c) Bob encrypts the message using Alice's private key. When Alice receives the message, she decrypts it using her public key. d) Each user places public key in a public register or other accessible file	1
	1.4) With respect to Key generation in DES, which of the following is False? a) For generation of 56 bit Cipher key, 8 extra parity bits are dropped from 64 bit Key. b) The round-key generator creates sixteen 48-bit keys out of a 56-bit cipher key. c) After Parity drop, the Key is divided into two parts of 28 bit each. d) Each part is shifted left by one bit for all rounds.	1
	1.5) Which of the following cannot be a component of Non-Feistel Cipher? a) Straight P Box b) S Box with equal no of inputs and outputs c) Compression P Box d) EXOR operation	1

	1.6) Which figure implements confidentiality and authentication? a)  b)  c)  d) 	1
	1.7) In MD5, the process block divides the 512 bits into _____ sub blocks. a) 16 b) 24 c) 32 d) 84	1
	1.8) MACs are also called as a) Testword b) Checkword c) Testbits d) None of the above	1
	1.9) Which mode of operation is used in the DAA? a) Output feedback mode b) Electronic code block mode c) Cipher block chaining mode d) Cipher feedback mode	1
	1.10) How many bits need to be padded for input message length of 2590 bits in SHA-512 ? a) 454 b) 354 c) 128 d) Can't be determined	1
Q.2 a)	Use the Playfair Cipher to encipher the message “The key is hidden under the door pad”. The secret key can be made by filling the first and part of the second row with the word “DEMOCRAT” and filling the rest of the matrix with the remaining alphabets.	5

Q.2 b)	Illustrate the Working of DES function with suitable diagrams. OR Explain the Implementation of Confusion and Diffusion in Modern Block Ciphers using suitable diagrams.	5
Q. 3	a) Define Cryptographic Hash Function? What are the requirements of Cryptographic Hash Function? b) List the difference between MD5 and SHA-1. OR Demonstrate the operation of SHA-512 with suitable structure.	5 5 10